Review

- What could cause error / Noise in the image data?
- Ways to assess the image quality
- Population,
- Sample, Biased Sample. Sampling Error
- Histogram
- Univariate Image Stats

Image Processing Mathematical Notation

- Notation is used to describe the mathematical operations applied to the digital remote sensor data:
  - i = a row (or line) in the imagery
  - j = a column (or sample) in the imagery
  - k = a band of imagery
  - l = another band of imagery
  - n = total number of pixels in an array
  - BV_{ijk} = brightness value in a row i, column j, of band k
  - BV_{ik} = ith brightness value in band k

Image Processing Mathematical Notation

- BV_{il} = *i*th brightness value in band *l*
- $\min_k$ = minimum value of band *k*
- $\max_k$ = maximum value of band *k*
- range_k = range of actual brightness values in band *k*
- quant_k = quantization level of band *k* (e.g., 2^8 = 0 to 255;
- μ_k = mean of band *k*
- var_k = variance of band *k*
- σ_k = standard deviation of band *k*

Image Processing Mathematical Notation

- skewness_k = skewness of a band k distribution
- cov_{kl} = covariance between pixel values in two bands, k and l
- r_{kl} = correlation between pixel values in two bands, k and l

Remote Sensing
Multivariate Statistics

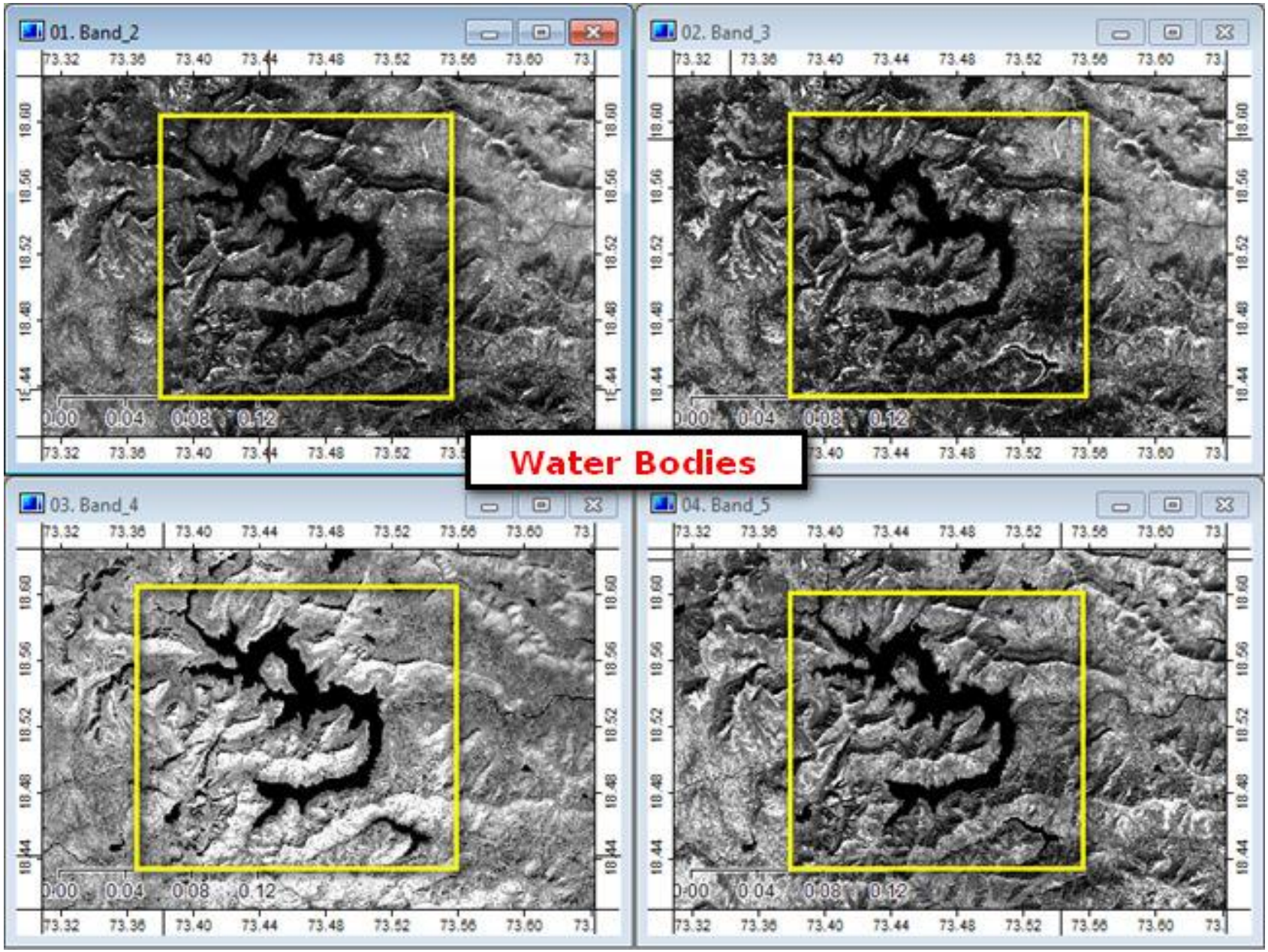
Remote Sensing Multivariate Statistics

- Different remote-sensing-derived spectral measurements for each pixel often change together in some predictable fashion
- If there is no relationship between the brightness value in one band and that of another for a given pixel, the values are mutually independent; that is,
  - an increase or decrease in one band's brightness value is not accompanied by a predictable change in another band's brightness value

Remote Sensing Multivariate Statistics

- It is useful to compute *multivariate* statistical measures such as *Covariance* and *Correlation* among the several bands to determine how the measurements **co-vary**
- Variance–covariance and correlation matrices are used in remote sensing
  - To find redundant data
  - Bands selection

Different objects responds differently at different wavelengths-Responses of objects change in different bands



Remote Sensing Multivariate Statistics

- **To calculate covariance between two bands, first compute the *corrected sum of products* (SP) defined by the equation:**

$$SP_{kl} = \sum_{i=1}^{n} \left(BV_{ik} - \mu_k\right)\left(BV_{il} - \mu_l\right)$$

- **Covariance: between brightness values in bands k and l,**

$$\text{COV}_{kl} = \frac{SP_{kl}}{n-1}$$

Covariance

- Large negative values indicate a strong negative correlation
- Large positive values show a clear positive relation
- Covariance values near zero indicate a weak or no correlation

Format of a Variance-Covariance Matrix

	Band 1 (green)	Band 2 (red)	Band 3 (near-infrared)	Band 4 (near-infrared)
Band 1	**var**	$cov_{1,2}$	$cov_{1,3}$	$cov_{1,4}$
Band 2	$cov_{2,1}$	**var**	$cov_{2,3}$	$cov_{2,4}$
Band 3	$cov_{3,1}$	$cov_{3,2}$	**var**	$cov_{3,4}$
Band 4	$cov_{4,1}$	$cov_{4,2}$	$cov_{4,3}$	**var**

Variance-Covariance Matrix of the Sample Data

	Band 1 (green)	**Band 2 (red)**	**Band 3 (near-infrared)**	**Band 4 (near-infrared)**
Band 1	**562.25**	-	-	-
Band 2	135	**264.80**	-	-
Band 3	718.75	275.25	**1007.50**	-
Band 4	537.50	64	663.75	**570**

Correlation between Multiple Bands of Remotely Sensed Data

- ***Correlation coefficient (r)*** **is commonly used to estimate the degree of interrelation (similarity) between variables**
- **The correlation between two bands of remotely sensed data, r_{kl}, is the ratio of their covariance (cov_{kl}) to the product of their standard deviations (σ_k and σ_l)**

$$r_{kl} = \frac{\mathrm{COV}_{kl}}{\sigma_k \sigma_l}$$

Correlation Coefficients

- Are normalized covariance values
- Correlation coefficient ranges from - 1 to +1
- A correlation close to +1 indicates a direct relationship between the two bands.
  - This suggests that if the reflectance (DN) of a pixel in one band is known, the reflectance of that pixel in the other band can be derived or estimated
- A correlation at -1 indicates an inverse relationship between the reflectance values of one band and the reflectance values in the other one

Correlation Matrix of the Sample Data

	Band 1 (green)	Band 2 (red)	Band 3 (near-infrared)	Band 4 (near-infrared)
Band 1	-	-	-	-
Band 2	0.92	-	-	-
Band 3	0.95	0.97	-	-
Band 4	0.34	0.16	0.57	-

Correlation Matrix

Band	Band 1	Band 2	Band 3	Band 4	Band 5	Band 6	Band 7
1	1	0.96	0.95	0.43	0.57	0.72	0.55
2	0.96	1	0.96	0.48	0.62	0.76	0.57
3	0.95	0.96	1	0.57	0.72	0.84	0.65
4	0.43	0.48	0.57	1	0.89	0.78	0.69
5	0.57	0.62	0.72	0.89	1	0.95	0.79
6	0.72	0.76	0.84	0.78	0.95	1	0.81
7	0.55	0.57	0.65	0.69	0.79	0.81	1

